LIVE MEMORY ACQUISITION AND ANALYSIS TOOLKIT

A PROJECT REPORT

Mini Project (01CE0609)

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A.Y. 2025-26

CERTIFICATE

This is to certify that the project report submitted along with the project entitled Live Memory Acquisition and Analysis Toolkit has been carried out by Vatsal Kakkad (92410103014) under my guidance in partial fulfillment for the degree of Bachelor of Technology in Computer Engineering, 6th Semester of Marwadi University, Rajkot during the academic year 2025-26.

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DECLARATION

We hereby declare that the **Mini Project (01CE0609)** report submitted along with the Project entitled Live Memory Acquisition Analysis Toolkit submitted in partial fulfilment for the degree of Bachelor of Technology in Computer Engineering to Marwadi University, Rajkot, is a bonafide record of original project work carried out by me / us at Marwadi University under the supervision of Prof. Parth Parmar and that no part of this report has been directly copied from any students' reports or taken from any other source, without providing due reference.

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Acknowledgement

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We would like to appreciate the efforts and cooperation of all group members, whose teamwork and dedication made this SRS document possible.

Abstract

The Memory Acquisition and Analysis Toolkit is a digital forensic system designed to capture, transfer, and analyze live system memory (RAM) to support cybersecurity investigations. RAM contains important volatile data such as running processes, active network connections, user sessions, and traces of malicious activities, which are lost once the system is shut down. Therefore, capturing this data in real time is essential for effective forensic analysis. The system focuses on acquiring memory from a Windows-based environment using tools like DumpIt or WinPmem, allowing investigators to collect data without interrupting system operations. The captured memory dump is then transferred to a Kali Linux environment for analysis using forensic frameworks such as Volatility and Rekall. The main objective of this system is to provide a structured and reliable approach for extracting useful information from memory, including system activities and potential malicious artifacts. The toolkit helps identify suspicious behavior, detect hidden or fileless malware, and analyze evidence not found on the hard disk. It also ensures an organized workflow for memory acquisition, analysis, and report generation, improving the efficiency and accuracy of digital forensic investigations.

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Abbreviations

RAM	**Random Access Memory**
ER	**Entity Relationship**
DFD	**Data Flow Diagram**
UML	**Unified Modeling Language**
OS	**Operating System**
GUI	**Graphical User Interface**
DMA	**Direct Memory Access**
VM	**Virtual Machine**
HTTP	**Hyper Text Transfer Protocol**
CPU	**Central Processing Unit**
DLL	**Dynamic Link Library**
FTK	**Forensic Toolkit**
PERT	**Program Evaluation and Review Technique**
WinPmem	**Windows Physical Memory**
DD	**Data Despritor**

CHAPTER 1

INTRODUCTION TO PROJECT AND PROJECT MANAGEMENT

1.1 PROJECT SUMMARY

The Memory Acquisition and Analysis Toolkit is a digital forensic project developed to capture and analyze live system memory (RAM) for cybersecurity investigations. The project focuses on collecting volatile data from a running Windows system using tools like DumpIt or WinPmem without shutting down the system. This memory data contains important information such as running processes, network connections, and possible malicious activities.

After capturing the memory dump, it is transferred to a Kali Linux environment where it is analyzed using forensic tools such as Volatility and Rekall. The analysis helps in identifying suspicious processes, detecting hidden or fileless malware, and extracting useful forensic evidence that may not be available on the hard disk.

The main goal of this project is to provide a structured workflow for memory acquisition, analysis, and report generation. It improves the efficiency and accuracy of digital forensic investigations and helps in detecting advanced cyber threats effectively.

1.2 PURPOSE

The purpose of the Memory Acquisition and Analysis Toolkit is to provide a systematic and efficient solution for capturing, transferring, and analyzing live system memory (RAM) to support digital forensic investigations and cybersecurity analysis. In many cases, important information such as running processes, active network connections, user activities, and traces of malicious programs exist only in volatile memory and are lost once the system is shut down. Therefore, this system is designed to acquire memory data from a running

Windows system without interrupting its operation, ensuring that critical evidence is preserved in real time.

The toolkit also enables the analysis of the captured memory dump using forensic frameworks in a Kali Linux environment. It helps investigators extract useful information, identify suspicious behavior, detect hidden or fileless malware, and analyze system-level activities that may not be available on the hard disk. Additionally, the system provides a structured workflow for memory acquisition, analysis, and report generation, making the investigation process more organized and reliable. Overall, the purpose of this project is to enhance the accuracy, efficiency, and effectiveness of digital forensic investigations by providing a practical and easy-to-use memory analysis solution.

1.3 OBJECTIVE

The main objective of the Memory Acquisition and Analysis Toolkit is to develop a system that can capture live system memory (RAM) from a running Windows machine without shutting it down. This ensures that important volatile data such as running processes, network connections, and system activities are preserved for forensic investigation. The project also aims to provide a reliable method for transferring the captured memory dump to a Kali Linux environment for further analysis.

Another objective of this system is to analyze the acquired memory using forensic tools such as Volatility and Rekall. Through this analysis, the system helps in extracting useful information like process details, network activity, and potential malicious artifacts. It also focuses on identifying suspicious behavior, detecting hidden or fileless malware, and uncovering threats that may not be visible in traditional disk-based analysis.

Additionally, the project aims to establish a structured workflow for memory acquisition, analysis, and report generation. This improves the efficiency, accuracy, and reliability of digital forensic investigations and assists investigators in effectively detecting cyber threats and preserving critical digital evidence.

1.4 SCOPE

The Memory Acquisition and Analysis Toolkit is developed to support digital forensic investigations by capturing and analyzing live system memory (RAM) from a Windows-based environment. The system enables investigators to acquire volatile memory data from a running system without shutting it down, ensuring that critical information such as running processes, active network connections, user sessions, and system-level activities are preserved. It supports the creation of memory dump files using external acquisition tools and facilitates the secure transfer of these files to a Kali Linux environment for further analysis. Using forensic frameworks such as Volatility and Rekall, the system allows the extraction of valuable forensic artifacts, including process lists, DLL information, network details, and potential indicators of compromise. The toolkit helps investigators identify suspicious behavior, detect hidden or fileless malware, and analyze system activities that are not visible through traditional disk-based forensics. Additionally, it provides a structured workflow for memory acquisition, analysis, and report generation, making the investigation process more organized, efficient, and reliable.

However, the scope of this system is limited in several aspects. The toolkit primarily focuses on live memory acquisition and analysis, and does not perform full disk forensics, file recovery, or detailed analysis of permanent storage devices. Although the system uses both Windows and Kali Linux environments, memory acquisition is performed only on Windows systems, while Kali Linux is used specifically for analysis purposes. The toolkit relies on external tools such as DumpIt, Volatility, and Rekall, and does not internally implement memory acquisition or analysis algorithms. Additionally, the system does not provide real-time monitoring or automatic threat prevention; instead, it is designed for post-acquisition forensic analysis of captured memory data. The accuracy of the results depends on the quality and completeness of the acquired memory dump and the correct usage of forensic tools. Furthermore, users are required to have basic technical knowledge of command-line operations and digital forensic concepts to effectively use the system and interpret the analysis results.

1.5 TEHNOLOGY AND LITERATURE REVIEW

Table 1.1 Literature Review With Technology-I

Paper Name	Authors	Tools Used	Technology	Advantages	Disadvantages	Novelty
Memory Forensic: Acquisition and Analysis of Memory	Mital Parekh, Snehal Jani	Volatility, Autopsy, RAM Dump	Memory Forensics, RAM Analysis	Extracts volatile data, supports malware detection	Extracts volatile data, supports malware detection	Compares forensic tools for effective analysis
Survey of Memory Acquisition Techniques	Stefan Vömel, Felix Freiling	Volatility, MDD, WinHex	RAM Acquisition, DMA, Cold Boot	Captures memory evidence, detects hidden malware	Captures memory evidence, detects hidden malware	Provides evaluation model for techniques
Memory Forensics: Review of Acquisition and Analysis Techniques	Dr. Grant Osborne	Volatility, Memoryze,WinPMem, WinHex	DMA, Virtualization, Cold Boot	Extracts keys, detects malware, captures system state	May alter system state, hardware dependency	Shows RAM analysis for reconstructing system activity

Table 1.2 Literature Review With Technology-II

Paper Name	Authors	Tools Used	Technology	Advantages	Disadvantages	Novelty
Anti-Forensic Memory Acquisition	Johanns Stüttgen, Michael Cohen	WinPmem, FTK Imager, DumpIt	Hardware-based Memory Acquisition	Recovers hidden memory, works across systems	Slow process, complex implementation	Resistant to anti-forensic techniques
Scenario-Based Assessment of Memory Tools	Lisa Rzepka	Belkasoft, FTK Imager, WinPMem	Volatility3, Virtual Addressing	Compares tools, supports investigation analysis	Limited testing, some dumps unsupported	Evaluates tool reliability in scenarios
Acquisition & Analysis of RAM Evidence	Rajesh Kumar	FTK Imager Lite, Magnet Axiom, Wireshark	Live RAM Acquisition, Windows Forensics	Captures volatile evidence, useful for legal cases	Requires expertise, data loss after shutdown	Focus on real-time RAM evidence in cybercrime

Table 1.3 Literature Review With Technology-III

Paper Name	Authors	Tools Used	Technology	Advantages	Disadvantages	Novelty
Fileless Malware Threats using Memory Forensics	Ilker Kara	Volatility, FTK Imager, Wireshark	Memory Forensics, RAM Analysis	Detects hidden malware in RAM	Requires expertise, resource intensive	Highlights detection of fileless malware
Memory Analysis – Techniques and Tools	Nuno Santos	Volatility, Memoryze, Memparser	Volatility, Memoryze, Memparser	Identifies hidden processes, detects malware	Complex methods, may affect system	Complex methods, may affect system
Hardware-Based Memory Acquisition	Brian Carrier, Joe Grand	Tribble, dd, Debug Tools	Hardware-based RAM Acquisition, DMA	Reliable evidence, prevents tampering	Reliable evidence, prevents tampering	Requires hardware setup, not flexible

Table 1.4 Literature Review With Technology-IV

Paper Name	Authors	Tools Used	Technology	Advantages	Disadvantages	Novelty
Data Acquisition for Firefox OS	Mandar Jadhav, K. K. Joshi	ADB, HexEditor MOBILedit	Firefox OS, Linux Kernel, ADB	Non-intrusive, uses open-source tools	Limited access, needs rooting	ADB-based method for Firefox OS
Memory Forensics: Tools & Techniques	Shreshtha Gaur, Rita Chhikara	Belkasoft, FTK Imager, wxHex	Live memory, Digital Forensics	Captures RAM data, detects malware	No single tool sufficient, complex	ML integration in memory forensics
Forensic Analysis of Physical Memory	Liming Cai, Jing Sha, Wei Qian	DumpIt, Volatility	Memory dump, Windows OS	Extracts processes, supports live analysis	Tools may alter memory, setup needed	Practical Windows memory analysis

Table 1.5 Literature Review With Technology-V

Paper Name	Authors	Tools Used	Technology	Advantages	Disadvantages	Novelty
Warm Boot Memory Acquisition	Timothy Vidas	Afterlife, memextract	USB-based, BIOS-level memory capture	No driver needed, portable	Limited memory, data overwrite risk	Captures RAM after reboot
Volatile Memory Forensic Analysis	Rushita Dave	Belkasoft, FTK, WinPmem	RAM dump, hashing, live analysis	Recovers passwords, detects malware	Data lost after shutdown, complex	Extracts artifacts from RAM
Physical Memory Analysis (Windows)	Shijin Chandran	Kernel driver, user-mode tools	Windows kernel, memory access	Efficient, supports multiple versions	Needs driver, complex setup	Kernel-based RAM + network data extraction

The Memory Acquisition and Analysis Toolkit uses a combination of operating systems, forensic tools, and programming support to perform memory acquisition and analysis. The primary platform for memory acquisition is a Windows-based system, where tools such as DumpIt or WinPmem are used to capture live memory (RAM) without shutting down the system. These tools are widely used in digital forensics for collecting volatile data safely and efficiently.

For memory analysis, the system uses a Kali Linux environment, which is a popular platform for cybersecurity and forensic analysis. Tools such as Volatility and Rekall are used to analyze the memory dump. These frameworks provide various plugins to extract information like running processes, network connections, registry data, and possible malicious activities. Python is used as the supporting language for running these tools and managing analysis tasks. Additionally, virtualization software such as VirtualBox or VMware is used to create and manage the Kali Linux environment on the host system.

1.6 PROJECT PLANNING AND SCHEDULING

1.6.1 Project Development Approach and Justification

The Memory Acquisition and Analysis Toolkit follows a structured and practical approach for capturing and analyzing live system memory. The development process begins with setting up two environments: a Windows system for memory acquisition and a Kali Linux system for analysis. Memory acquisition tools such as DumpIt or WinPmem are configured on the Windows system to capture live RAM data without shutting down the system, ensuring that volatile information is preserved. Once the memory dump is created, it is transferred securely to the Kali Linux environment using network sharing or other transfer methods. In the analysis phase, forensic tools such as Volatility and Rekall are used to extract useful information, including running processes, network connections, and potential malicious activities. The final step involves organizing the extracted data into a structured report for investigation purposes.

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the analysis phase, forensic tools such as Volatility and Rekall are used to extract useful information, including running processes, network connections, and potential malicious activities. The final step involves organizing the extracted data into a structured report for investigation purposes.

1.6.2 Project Effort and Time ,Cost Estimation

The Memory Acquisition and Analysis Toolkit project was carried out from 21 January 2026 to 9 April 2026. The total duration of the project is 79 days. The project was completed by a team of 2 persons, working collaboratively on different phases such as planning, design, implementation, testing, and documentation.

For this project, a team of 2 members is considered sufficient, as the system involves setup, analysis, and documentation tasks that can be efficiently handled by a small team. The standard working time is assumed to be 6 hours per day per person.

Effort is calculated using the standard formula:

Effort (Person-Hours) = Number of Persons × Working Hours per Day × Number of Days

Effort = 2 × 6 × 79 = 948 Person-Hours

For Time distribution

Planning & Requirement Analysis → 10 days

Design (Architecture, ER, DFD, UML) → 15 days

Implementation (Setup + Memory Acquisition + Analysis) → 34 days

Testing & Debugging → 10 days

Documentation & Presentation → 10 days

For Cost Estimation

Cost = Total Person-Hours × Cost per Hour

Assuming the cost per hour is ₹100 per hour:

Total Cost = 948 × 100 = ₹94,800

1.6.3 Roles and Responsibilities

The development of the Memory Acquisition and Analysis Toolkit was carried out by the following team members, each assigned specific roles and responsibilities to ensure efficient project execution.

Vatsal Kakkad (Project Developer & Analyst):

Responsible for the overall development and implementation of the system. This includes setting up the Windows and Kali Linux environments, performing memory acquisition using tools like DumpIt, and analyzing memory dumps using Volatility and Rekall. Also responsible for designing system architecture and preparing diagrams such as ER, DFD, and workflow diagrams.

Vatsal Kakkad (System Tester & Support):

Responsible for testing the system to ensure correct functionality. This includes verifying memory acquisition, validating analysis results, identifying errors, and assisting in troubleshooting. Also supports system setup and execution of forensic tools.

Hemanshi Jalu (Documentation & Presentation):

Responsible for preparing the project report, maintaining proper documentation, and creating presentation materials. Ensures that all project details are clearly documented and presented effectively during evaluation.

1.6.4 Group Dependencies

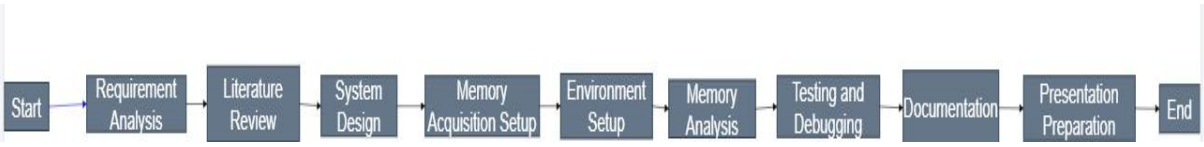
The development of the Memory Acquisition and Analysis Toolkit involves coordination and dependency among team members to ensure smooth execution of tasks. Since the project consists of multiple phases such as memory acquisition, analysis, testing, and documentation, each task depends on the successful completion of the previous stage.

The memory acquisition process depends on proper system setup and configuration of tools on the Windows environment. Once the memory dump is generated, the analysis phase depends on the successful transfer of the dump file to the Kali Linux system. The analysis using tools such as Volatility and Rekall relies on the availability of a valid and complete memory dump. Therefore, any issue in acquisition or transfer can affect the accuracy of the analysis.

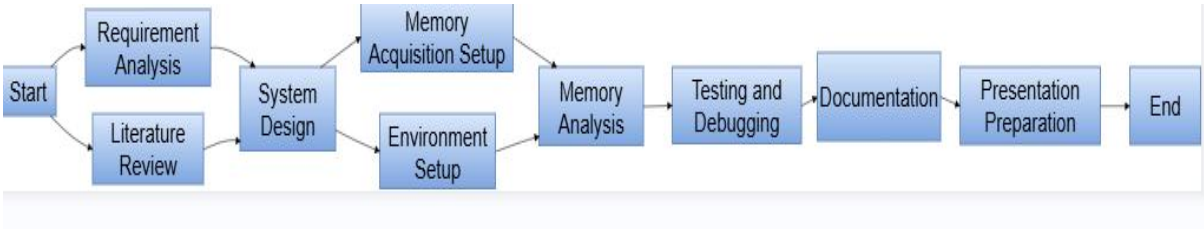
The testing phase depends on the outputs generated during the analysis stage. The tester verifies whether the extracted information is correct and meaningful. Any errors identified during testing require coordination with the developer to fix issues and improve system performance. Similarly, the documentation and presentation tasks depend on the results and findings obtained from all previous phases of the project.

Overall, effective communication and coordination among team members are essential, as each role depends on the output of another. Proper task management and timely completion of each phase ensure the successful development and execution of the project.

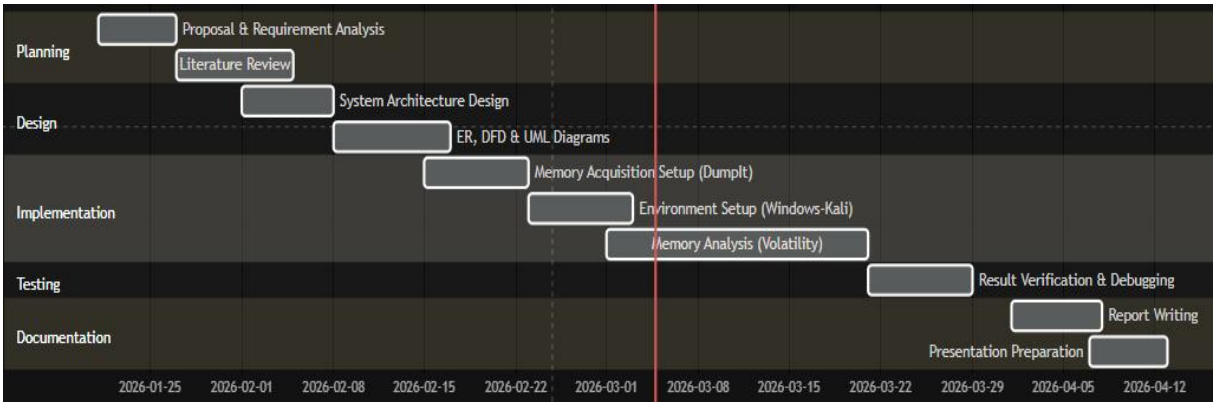
1.7 PROJECT SEHEDULING



1.1 PERT Chart



1.2 Network Chart



1.3 Gantt Chart

CHAPTER 2

SYSTEM ANALYSIS

2.1 STUDY OF CURRENT SYSTEM

In the current scenario, digital forensic investigations mainly focus on analyzing data stored on hard disks and other permanent storage devices. Traditional forensic methods rely heavily on disk imaging and file system analysis to detect malicious activities. However, these methods often overlook volatile memory (RAM), which contains critical real-time information such as running processes, active network connections, user sessions, and temporary data.

In the current scenario, digital forensic investigations mainly focus on analyzing data stored on hard disks and other permanent storage devices. Traditional forensic methods rely heavily on disk imaging and file system analysis to detect malicious activities. However, these methods often overlook volatile memory (RAM), which contains critical real-time information such as running processes, active network connections, user sessions, and temporary data.

Existing systems for memory analysis are mostly tool-based and not integrated into a single structured workflow. Investigators use separate tools like DumpIt, Volatility, and Rekall individually, which requires manual effort to perform memory acquisition, transfer, and analysis. This lack of integration makes the process time-consuming and complex, especially for beginners. Additionally, improper handling during memory acquisition can lead to data loss or contamination, affecting the reliability of the investigation.

2.2 PROBLEM AND CURRENT WEAKNESS IN THE SYSTEM

In the current digital forensic process, most investigations primarily focus on analyzing data stored on hard disks, often neglecting volatile memory (RAM). This creates a major problem, as important real-time information such as running processes, active network connections,

user sessions, and traces of malicious activities reside only in memory and are lost when the system is shut down. As a result, critical evidence may not be captured, reducing the effectiveness of the investigation.

In the current digital forensic process, most investigations primarily focus on analyzing data stored on hard disks, often neglecting volatile memory (RAM). This creates a major problem, as important real-time information such as running processes, active network connections, user sessions, and traces of malicious activities reside only in memory and are lost when the system is shut down. As a result, critical evidence may not be captured, reducing the effectiveness of the investigation.

Another major weakness is the lack of a structured and integrated system for memory acquisition and analysis. Investigators are required to use multiple independent tools such as DumpIt, Volatility, and Rekall separately, which makes the process complex and time-consuming. There is no unified workflow to guide users from memory acquisition to analysis and report generation, leading to inefficiency and possible errors.

Additionally, many existing approaches require a high level of technical expertise. Beginners may find it difficult to correctly acquire memory, transfer dump files, and interpret analysis results. Improper handling during acquisition can lead to data corruption or incomplete memory dumps, which affects the reliability of the investigation.

2.3 REQUIREMENT OF NEW SYSTEM

The increasing complexity of cyber threats and the limitations of existing forensic methods create the need for a more advanced and structured system for memory analysis. The current approaches often fail to capture and analyze volatile memory effectively, leading to loss of critical evidence. Therefore, a new system is required that can efficiently handle live memory acquisition and analysis in a reliable and organized manner.

The proposed system should be capable of capturing live system memory (RAM) from a running Windows environment without shutting it down, ensuring that important volatile data such as running processes, network connections, and system activities are preserved. It should provide a seamless method for transferring the acquired memory dump to a Kali Linux environment for further analysis.

The system must support the use of advanced forensic tools such as Volatility and Rekall to analyze memory data and extract useful information. It should be able to identify suspicious processes, detect hidden or fileless malware, and uncover potential security threats that are not visible through traditional disk-based analysis.

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Overall, the requirement of the new system is to enhance the efficiency, accuracy, and reliability of digital forensic investigations by providing a complete and practical solution for handling volatile memory data.

2.4 SYSTEM FEASIBILITY

2.4.1 Does the System Contribute to the Overall Objective of the Organization?

Yes, the Memory Acquisition and Analysis Toolkit significantly contributes to the overall objectives of an organization, especially in terms of cybersecurity, data protection, and digital investigation capabilities. Organizations aim to ensure the security and integrity of their systems and data, and this system helps achieve that by enabling effective detection and analysis of security threats present in volatile memory.

The system supports the identification of suspicious activities, hidden processes, and advanced threats such as fileless malware, which are often not detectable through traditional methods. By providing the ability to capture and analyze live memory, it helps organizations respond quickly to security incidents and gather critical evidence for investigation and decision-making.

Additionally, the toolkit improves the efficiency and accuracy of forensic investigations by offering a structured workflow for memory acquisition, analysis, and reporting. This reduces investigation time and enhances reliability, which aligns with organizational goals of maintaining secure and well-monitored systems.

2.4.2 Can the System be Implemented Using the Current Technology and within the given Cost and Schedule Constraints.

Yes, the Memory Acquisition and Analysis Toolkit can be successfully implemented using current technology within the given cost and schedule constraints. The system relies on widely available and well-established tools such as DumpIt or WinPmem for memory acquisition and Volatility and Rekall for memory analysis. These tools are compatible with modern systems and do not require any specialized or expensive hardware, making the implementation technically feasible.

The project utilizes a Windows environment for memory acquisition and a Kali Linux environment for analysis, both of which can be easily set up using virtualization software such as VirtualBox or VMware. This eliminates the need for additional physical systems and reduces overall cost. The technologies involved are open-source or freely available, which further ensures that the system can be developed within a low budget.

The project utilizes a Windows environment for memory acquisition and a Kali Linux environment for analysis, both of which can be easily set up using virtualization software such as VirtualBox or VMware. This eliminates the need for additional physical systems and reduces overall cost. The technologies involved are open-source or freely available, which further ensures that the system can be developed within a low budget.

Overall, the system is both technically and economically feasible, and it can be implemented effectively using current technologies within the planned cost and schedule constraints.

2.4.3 Can the System be integrated with other system which are already in place?

Yes, the Memory Acquisition and Analysis Toolkit can be integrated with other systems that are already in place within an organization. The toolkit is designed to work with standard operating systems such as Windows for memory acquisition and Kali Linux for analysis, making it compatible with most existing IT infrastructures. It uses widely accepted forensic tools such as DumpIt, Volatility, and Rekall, which can easily be incorporated into existing digital forensic workflows.

The system can be integrated with incident response systems, security monitoring tools, and

forensic investigation processes to enhance overall cybersecurity capabilities. For example, memory dumps collected using this toolkit can be used alongside other forensic data such as disk images and network logs to provide a more comprehensive analysis of security incidents.

Additionally, since the system uses common file formats (such as memory dump files), it can be easily shared and processed by other forensic tools and platforms. The use of virtualization also allows the system to be deployed without affecting existing systems.

2.5 ACTIVITY /PROCESS IN NEW SYSTEM /PROPOSED SYSTEM

The Memory Acquisition and Analysis Toolkit follows a structured and systematic process to capture and analyze live system memory for digital forensic investigations. The process begins with setting up the required environments, including a Windows system for memory acquisition and a Kali Linux system for analysis. Necessary tools such as DumpIt or WinPmem are configured on the Windows system, while Volatility and Rekall are installed on the Kali Linux system.

The first activity involves live memory acquisition, where the system captures the RAM data from the running Windows machine without shutting it down. This ensures that volatile information such as running processes, network connections, and system activities is preserved. Once the memory dump file is generated, the next step is the secure transfer of this file to the Kali Linux environment using network sharing or other transfer methods.

After the transfer, the memory analysis process is performed using forensic tools like Volatility and Rekall. These tools analyze the memory dump to extract useful information such as process lists, network details, loaded modules, and potential malicious artifacts. The system then proceeds to the result interpretation phase, where the extracted data is examined to identify suspicious activities, hidden processes, or signs of malware.

Finally, the system performs report generation, where all findings are organized into a structured format for documentation and investigation purposes. This complete workflow ensures an efficient, reliable, and organized approach to memory acquisition and analysis, improving the overall effectiveness of digital forensic investigations.

2.6 FEATURES OF NEW SYSTEM / PURPOSED SYSTEM

The Memory Acquisition and Analysis Toolkit provides several important features that improve the efficiency and effectiveness of digital forensic investigations. One of the key features is live memory acquisition, which allows the system to capture RAM data from a running Windows system without shutting it down, ensuring that valuable volatile information is preserved.

Another important feature is cross-platform operation, where memory acquisition is performed on a Windows system and analysis is carried out in a Kali Linux environment. This separation enhances flexibility and allows the use of powerful forensic tools. The system supports advanced memory analysis using frameworks such as Volatility and Rekall, which help extract detailed information like running processes, network connections, and system activities.

The toolkit also includes a structured workflow that integrates memory acquisition, transfer, analysis, and report generation, making the process organized and easy to follow. It provides the ability to detect hidden or fileless malware, which cannot be identified through traditional disk-based methods. Additionally, the system ensures efficient data handling and transfer, allowing secure movement of memory dump files between systems.

Another feature is report generation, where analyzed data is presented in a clear and structured format for investigation and documentation purposes. The system is also cost-effective and easy to implement, as it uses widely available tools and requires minimal hardware resources.

Overall, these features make the proposed system reliable, efficient, and suitable for modern digital forensic and cybersecurity applications

2.7 LIST OF MAIN MODULES /COMPONENTS /PROCESSES /TECHNIQUES OF NEW SYSTEM /PROPOSED SYSTEM

The Memory Acquisition and Analysis Toolkit is divided into several key modules and components, each responsible for a specific function in the overall system. These modules work together to provide a complete and structured workflow for memory acquisition and analysis.

Memory Acquisition Module:This module is responsible for capturing live system memory (RAM) from a running Windows system using tools such as DumpIt or WinPmem. It ensures that volatile data is collected without shutting down the system.

Data Transfer Module:This module handles the secure transfer of the memory dump file from the Windows system to the Kali Linux environment. It uses network sharing or other transfer methods to move the data safely.

Memory Analysis Module:This module performs analysis of the captured memory dump using forensic frameworks such as Volatility and Rekall. It extracts important information like running processes, network connections, and system activities.

Threat Detection Module:This module focuses on identifying suspicious activities, hidden processes, and fileless malware present in the memory. It helps in detecting potential security threats.

Result Interpretation Module:This module analyzes and organizes the extracted data to identify meaningful patterns and evidence useful for investigation.

Report Generation Module:This module generates a structured report based on the analysis results, which can be used for documentation and forensic investigation.

Technology used

1.Live Memory Acquisition

2.Memory Forensics Analysis

3.Process and Network Analysis

4.Malware Detection Techniques

5.Data Transfer Techniques

2.8 SELECTION OF HARDWARE /SOFTWARE /ALGORITHMS /METHODOLOGY / TECHNIQUES /APPROACHES AND JUSTIFICATION

The development of the Memory Acquisition and Analysis Toolkit requires careful selection of hardware, software, methodologies, and techniques to ensure efficient and reliable performance. The selected components are based on availability, compatibility, cost-effectiveness, and suitability for digital forensic investigations.

The hardware requirements include a standard computer system with sufficient RAM (minimum 8 GB recommended) and storage capacity to handle memory dump files. No specialized hardware is required, which makes the system cost-effective and easy to implement in most environments.

The software selection includes a Windows operating system for performing live memory acquisition and a Kali Linux environment for analysis. Tools such as DumpIt or WinPmem are used for capturing memory, while Volatility and Rekall are used for analysis. These tools are widely used, reliable, and mostly open-source, making them suitable for academic and practical use.

The system follows a structured methodology, where the process is divided into stages such as memory acquisition, data transfer, analysis, and report generation. This step-by-step approach ensures that each phase is executed correctly and the overall workflow remains organized and efficient.

The techniques used include live memory acquisition, memory forensics analysis, process and network analysis, and malware detection techniques. These techniques are essential for extracting meaningful information from volatile memory and identifying potential security threats.

The approach adopted is practical and aligns with real-world digital forensic practices. It uses existing tools and environments rather than developing new algorithms, which reduces complexity and development time. The approach is also flexible, allowing easy replication and modification based on requirements.

Overall, the selected hardware, software, methodology, and techniques are justified as they provide a reliable, cost-effective, and efficient solution for memory acquisition and analysis, making the system suitable for both academic and real-world applications.

CHAPTER 3

SYSTEM DESIGN

3.1 SYSTEM DESIGN AND METHODOLOGY

The Memory Acquisition and Analysis Toolkit is designed using a structured and modular approach to ensure efficient handling of memory acquisition and analysis. The system architecture consists of two main environments: a Windows-based system for live memory acquisition and a Kali Linux system for memory analysis.

The system workflow begins with the memory acquisition phase, where tools such as DumpIt or WinPmem are used to capture the RAM data from a running Windows system without shutting it down. The captured memory dump is then securely transferred to the Kali Linux environment through network sharing or other transfer methods. In the analysis phase, forensic frameworks such as Volatility and Rekall are used to extract important information like running processes, network connections, and system activities from the memory dump.

The methodology followed in this system is a step-by-step forensic approach, ensuring that each stage of the process is carried out systematically. The process includes acquisition, preservation, analysis, and reporting. During the analysis stage, various techniques such as process analysis, network analysis, and malware detection are applied to identify suspicious activities and potential threats. The final stage involves report generation, where all findings are documented in a structured format for investigation and evaluation purposes.

3.2 DATABASE DESIGN / DATA STRUCTURE DESIGN / PROCESS DESIGN / STRUCTURE DESIGN / ER DIAGRAM /DFD / UML

Database Design:

The database design of the Memory Acquisition and Analysis Toolkit focuses on storing and managing important information related to memory acquisition and forensic analysis in a structured and organized manner. The system maintains data such as user details, system information, memory dump files, and analysis results. The main entities in the database include User, System, Memory Dump, and Analysis Results. Each entity represents a key part

of the process, where the User performs operations on a System, generates a Memory Dump, and obtains Analysis Results. This structured storage helps in maintaining consistency and easy access to required data.

These entities are connected through relationships defined in the ER diagram, which shows how data flows within the system. For example, a user acquires memory from a system, which creates a memory dump that is further analyzed to produce results. The database design ensures data integrity, avoids redundancy, and supports efficient data retrieval. Overall, it provides a reliable structure for storing and managing forensic data, making the analysis process more effective and organized.

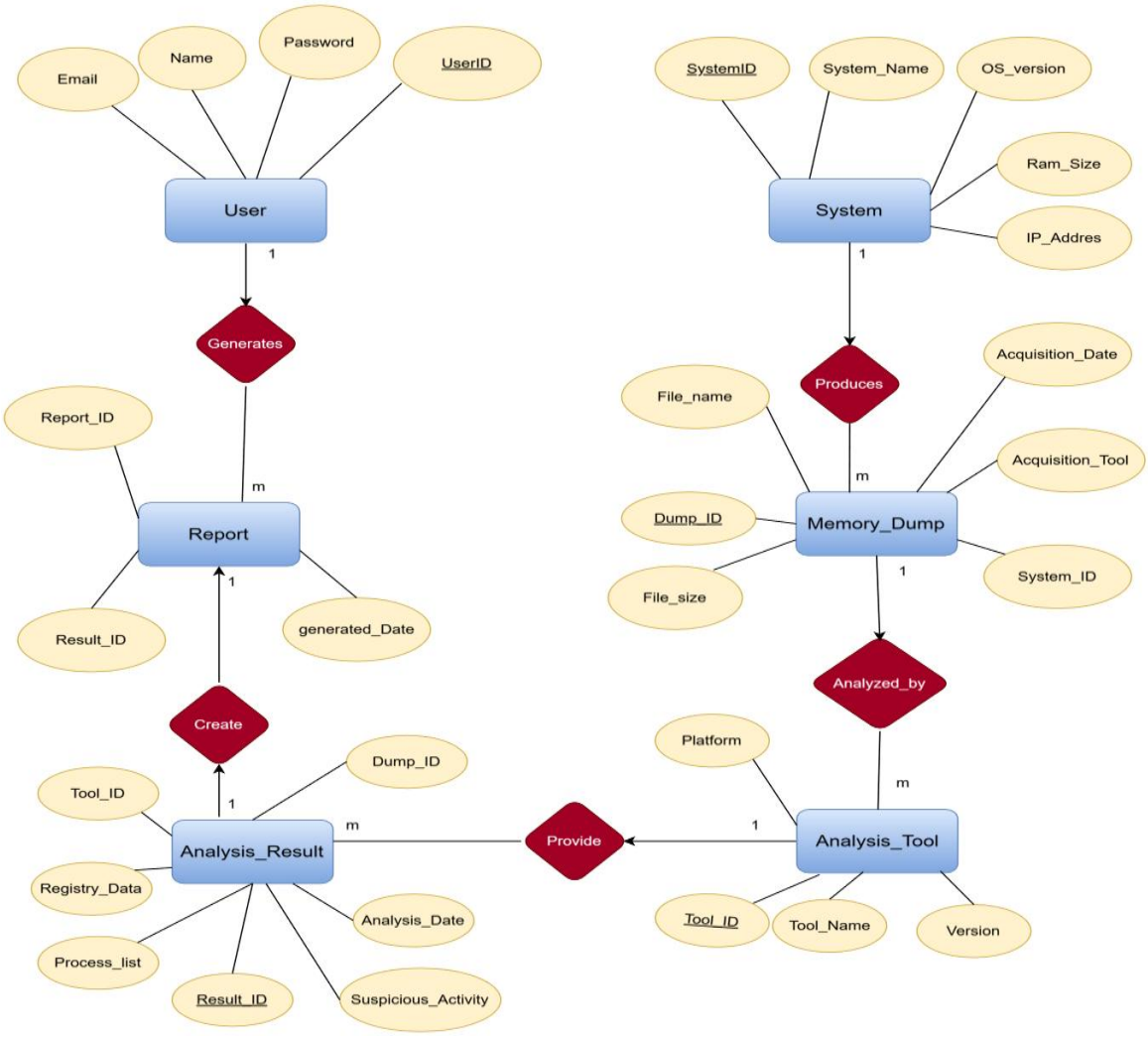

Fig 3.1 Er Diagram

Data Structure:

The data structure design of the Memory Acquisition and Analysis Toolkit defines how different types of data are organized, stored, and processed within the system. It includes handling of memory dump files, extracted process information, network details, and analysis outputs. The system uses structured formats to manage this data efficiently, ensuring that large memory dump files and extracted information can be processed without errors. Proper data organization helps in maintaining clarity and consistency during the analysis process.

The system structures data such as process lists, active network connections, and detected malware in an organized manner, making it easier to analyze and interpret results. These data structures support efficient searching, filtering, and reporting of forensic information. By using well-defined formats, the system ensures smooth data flow between different modules, reduces complexity, and improves overall performance and reliability of the memory analysis process.

Process Design:

The process design of the Memory Acquisition and Analysis Toolkit is represented using Data Flow Diagrams (DFD Level 0 and Level 1), which illustrate the overall workflow of the system. These diagrams provide a clear view of how the system operates, starting from memory acquisition to final report generation. The DFD Level 0 shows the high-level view of the system, including major processes such as memory capture, data transfer, analysis, and output generation, while DFD Level 1 provides a more detailed breakdown of these processes.

The process design clearly defines how data flows between different modules of the system. It shows how memory data is collected from the Windows system, transferred to the Kali Linux environment, and analyzed using forensic tools. The extracted information is then processed and converted into meaningful results. This structured flow ensures that each step is performed in a logical sequence, improving system efficiency, accuracy, and reliability in digital forensic investigations.

In addition, the process design ensures proper handling of data at each stage, including validation and error checking. During memory acquisition, the system ensures that the dump file is created successfully without corruption. During the transfer phase, it ensures that the file is securely moved without data loss. In the analysis phase, the system verifies outputs generated by forensic tools to maintain accuracy and consistency in results.

Furthermore, the process design supports a modular and flexible workflow, allowing each phase to be executed independently if required. This makes the system easy to maintain, update, and extend in the future. It also helps users understand the step-by-step operation of the system, making it easier to use and manage during forensic investigations.

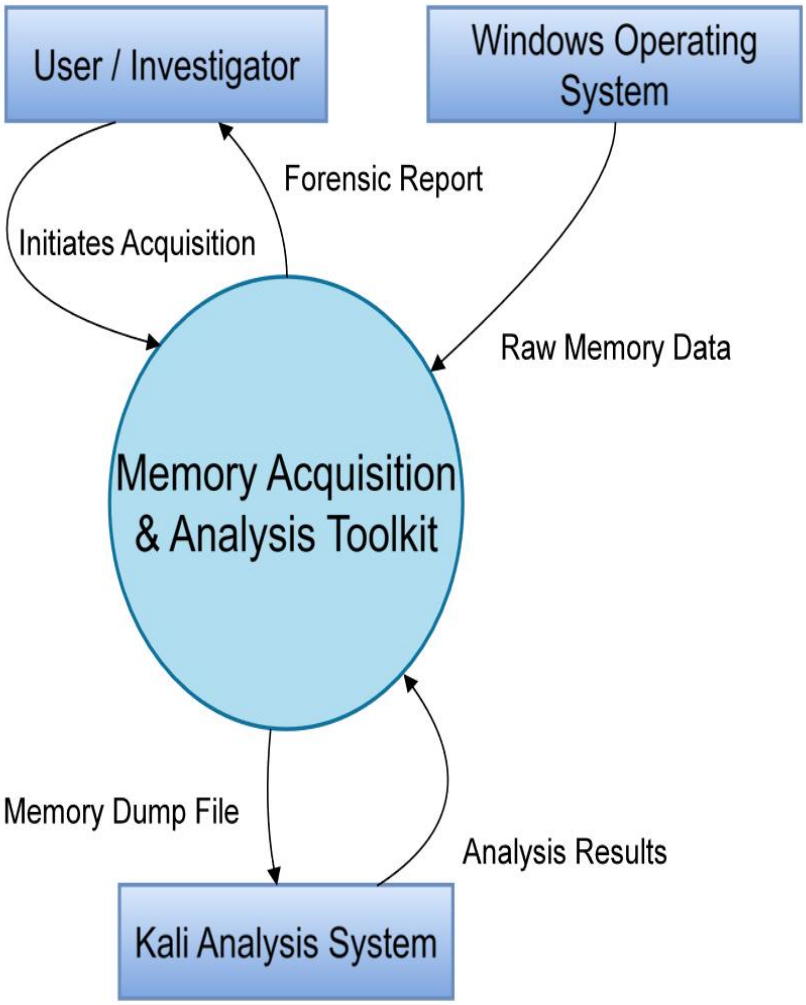

Fig 3.2 DFD Level 0 Diagram

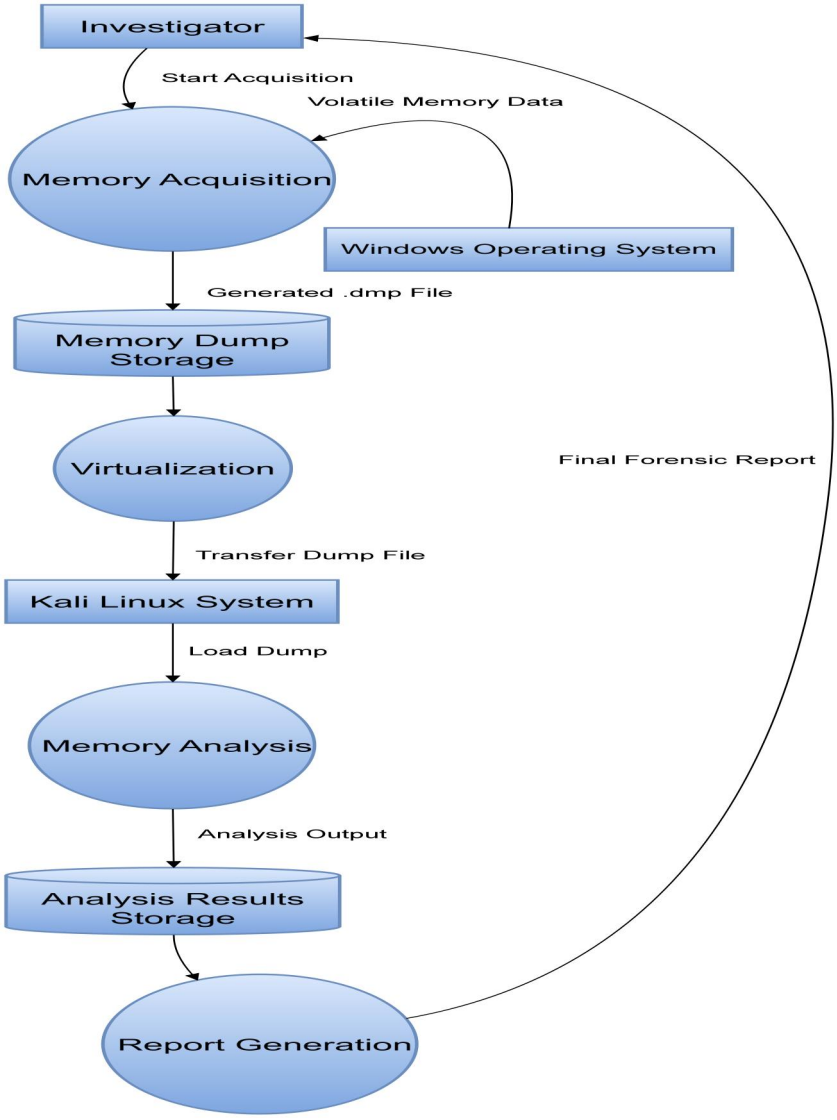

Fig 3.3 DFD Level 1 Diagram

Structure Design:

The structure design of the Memory Acquisition and Analysis Toolkit defines the overall organization and arrangement of system components and modules. The system is designed using a modular structure, where each module performs a specific function such as memory acquisition, data transfer, analysis, threat detection, and report generation. This modular approach improves system clarity, maintainability, and scalability, as each component can be developed and managed independently. The structure is also divided into two main environments: the Windows system for memory acquisition and the Kali Linux system for analysis.

The structure design is represented through the system architecture diagram, which shows how different components interact with each other. It illustrates the flow of data from the acquisition module to the analysis module and finally to the reporting module. This structured arrangement ensures smooth communication between modules and efficient processing of data. Overall, the design provides a clear framework for system operation, making it easier to understand, implement, and extend in future enhancements.

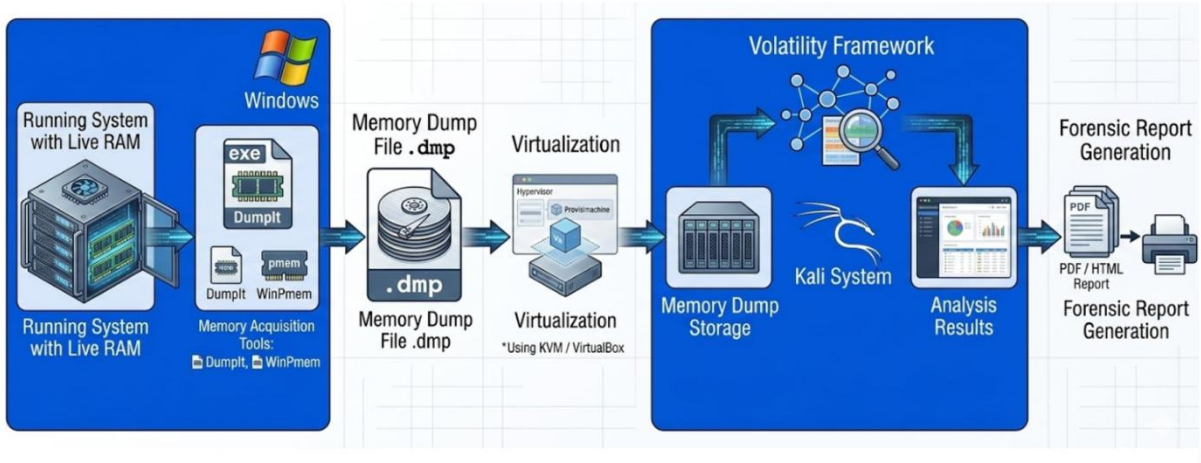

Fig 3.4 Architecture of Toolkit

UML Diagram:

The UML diagrams are used in the Memory Acquisition and Analysis Toolkit to represent the system behavior and interaction between different components. In this project, a sequence diagram is used to illustrate the step-by-step interaction between the user, system, and various modules during the memory acquisition and analysis process. It shows how the user initiates the process, how the system captures the memory, transfers the dump file, and performs analysis using forensic tools.

The UML diagram helps in understanding the flow of operations and the communication between different system components over time. It provides a clear visualization of how each step is executed in sequence, from data collection to result generation. This makes the system easier to understand, design, and implement, while also helping developers and users to identify the role of each component in the overall workflow.

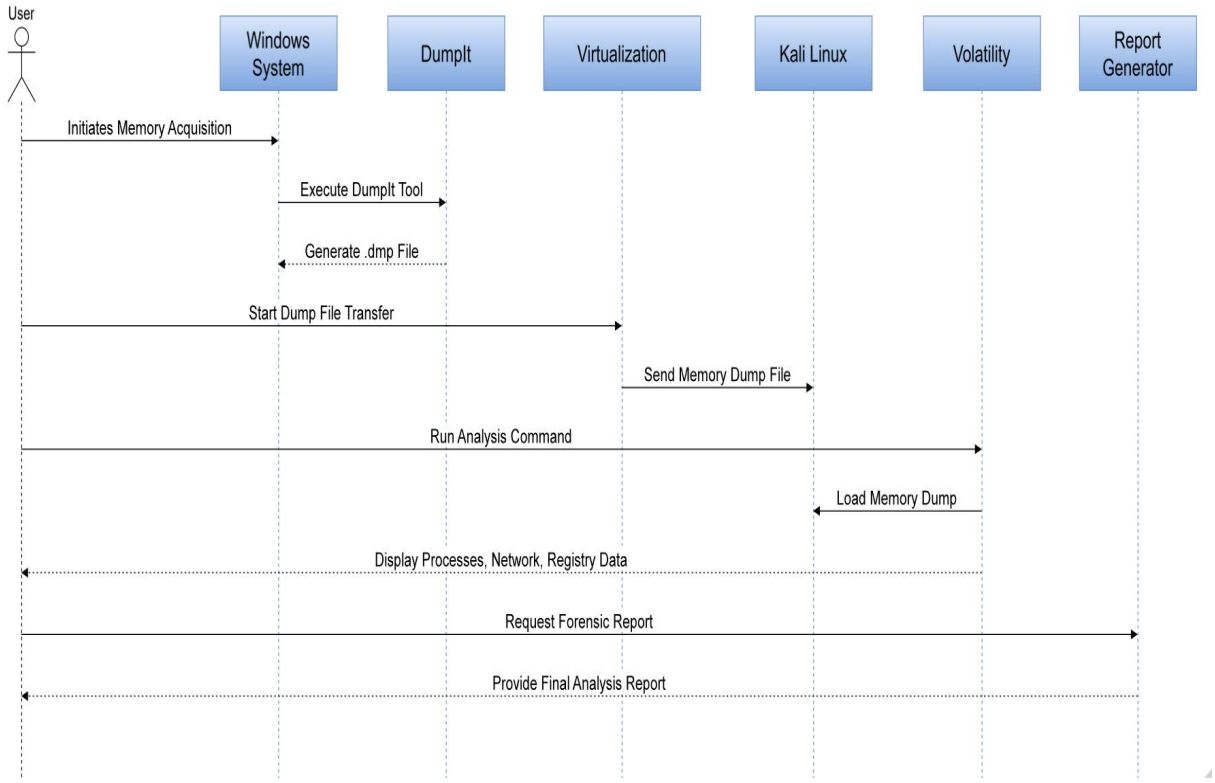

Fig 3.5 Sequence Diagram

CHAPTER 4

IMPLEMENTATION AND TESTING

4.1 IMPLEMENTATION PLATFORM / ENVIROMENT

The Memory Acquisition and Analysis Toolkit is implemented using a combination of Windows and Kali Linux environments to perform memory acquisition and analysis efficiently. The Windows operating system is used as the primary platform for capturing live memory, as it allows the use of tools such as DumpIt or WinPmem to acquire RAM data from a running system without shutting it down. This ensures that important volatile information is preserved during the acquisition process.

The analysis of the captured memory dump is performed in a Kali Linux environment, which provides a powerful platform for digital forensics and cybersecurity tasks. Tools such as Volatility and Rekall are installed and executed in this environment to extract and analyze data from the memory dump. The Kali Linux system can be set up using virtualization software such as VirtualBox or VMware, allowing it to run alongside the Windows system without requiring separate hardware.

The implementation environment also includes Python, which is used to run forensic tools and scripts required for memory analysis. A network-based file transfer method or shared folder is used to transfer the memory dump file from the Windows system to the Kali Linux system securely. The overall platform is designed to be cost-effective, easy to set up, and compatible with commonly available systems.

4.2 PROCESS /PROGRAM /TECHNOLOGY / MODULES SPECIFICATION

The Memory Acquisition and Analysis Toolkit is designed using a structured process, supported by specific programs, technologies, and well-defined modules. The system follows a step-by-step process that includes memory acquisition, data transfer, memory analysis,

result interpretation, and report generation. Each stage of the process is executed sequentially to ensure accurate handling of volatile memory data and effective forensic investigation.

The programs used in the system include memory acquisition tools such as DumpIt or WinPmem, and memory analysis frameworks such as Volatility and Rekall. These programs are executed through command-line interfaces and are responsible for capturing memory dumps and extracting useful forensic information. Python is used as a supporting technology to run analysis tools and scripts, making the process more flexible and efficient.

The technologies involved include Windows operating system for memory acquisition and Kali Linux for analysis. The system also uses virtualization technology (VirtualBox or VMware) to run multiple environments on a single machine. Additionally, networking techniques are used for transferring memory dump files between systems securely.

The system is divided into multiple modules, each responsible for a specific function. The Memory Acquisition Module captures RAM data, the Data Transfer Module handles secure file transfer, the Memory Analysis Module processes the dump file using forensic tools, the Threat Detection Module identifies suspicious activities, and the Report Generation Module prepares the final output. These modules work together to provide a complete and efficient workflow.

4.3 FINDING /RESULTS / OUTCOMES

The Memory Acquisition and Analysis Toolkit successfully demonstrates the ability to capture and analyze live system memory for digital forensic purposes. The system is able to generate a complete memory dump from a running Windows system without shutting it down, ensuring that important volatile data such as running processes, active network connections, and system activities are preserved. The acquired memory dump is then effectively transferred and analyzed in the Kali Linux environment using forensic tools.

The analysis results provide valuable insights into the system's internal state, including process lists, network details, and loaded modules. The system is capable of identifying suspicious or hidden processes and detecting potential security threats, including fileless malware that cannot be detected through traditional disk-based methods. This demonstrates

the effectiveness of memory forensics in uncovering advanced cyber threats.

The toolkit also ensures a structured workflow from acquisition to report generation, making the entire process organized and easy to follow. The results obtained are reliable and useful for forensic investigations, helping in evidence collection and analysis. Overall, the system achieves its objective of improving the efficiency, accuracy, and reliability of digital forensic analysis using live memory data.

4.4 RESULT ANALYSIS / COMPARISION / DELIBERATIONS

The results obtained from the Memory Acquisition and Analysis Toolkit demonstrate that live memory analysis provides deeper insights compared to traditional disk-based forensic methods. The system successfully extracts critical information such as running processes, active network connections, and system activities directly from RAM. This allows investigators to identify suspicious behavior and hidden processes that may not be visible in stored data. The analysis confirms that memory forensics is highly effective in detecting advanced threats, especially fileless malware.

In comparison to existing approaches that focus mainly on hard disk analysis, the proposed system offers a more comprehensive investigation by including volatile memory data. Traditional methods may miss temporary or in-memory activities, whereas this system captures real-time data before it is lost. Additionally, the use of tools like Volatility and Rekall provides more detailed and structured outputs, improving the accuracy and depth of analysis.

The deliberations from the project highlight that a structured workflow from memory acquisition to report generation significantly improves efficiency and reduces complexity. However, the accuracy of results depends on proper memory acquisition and correct usage of analysis tools. Overall, the system proves to be more effective, reliable, and suitable for modern digital forensic investigations compared to conventional methods.

4.5 TESTING PLAN / STRATEGY

The Memory Acquisition and Analysis Toolkit follows a systematic testing strategy to ensure that all components of the system function correctly and produce reliable results. The testing process is carried out in multiple stages, including unit testing, integration testing, and system testing. Each module, such as memory acquisition, data transfer, analysis, and report generation, is tested individually to verify its functionality and accuracy before being integrated into the complete system.

During unit testing, individual modules are tested to ensure they perform their specific tasks correctly. For example, the memory acquisition module is tested to verify that it successfully creates a valid memory dump file, while the analysis module is tested to ensure accurate extraction of processes and network information. In integration testing, the interaction between modules is tested, such as verifying that the memory dump file is correctly transferred and properly analyzed in the Kali Linux environment.

The system testing phase evaluates the complete workflow of the system, from memory acquisition to final report generation. The system is tested under different scenarios to check its performance, reliability, and error handling capabilities. Additionally, validation testing is performed to ensure that the results generated by the system are accurate and meaningful for forensic investigation. Overall, the testing strategy ensures that the system is stable, efficient, and capable of handling real-world forensic tasks effectively.

In addition, performance and usability testing are conducted to evaluate how the system handles large memory dump files and how easily users can operate the toolkit. The system is tested under different system configurations to ensure consistent performance. User feedback is also considered to improve usability and simplify the workflow. This ensures that the system is not only technically correct but also practical and user-friendly for real-world forensic investigations.

4.6 TEST RESULTS AND ANALYSIS

4.6.1 Test Cases

```
C:\Users\student\volatility3-stable>python vol.py -f MA202-55-20260325-020603.dmp windows.info
Volatility 3 Framework 2.27.0
Progress:  100.00               PDB scanning finished
Variable        Value

Kernel Base     0xf80322e00000
DTB     0x1ad000
Symbols file:///C:/Users/student/volatility3-stable/volatility3/symbols/windows/ntkrnlmp.pdb/B6121DA15DDCF625C8C7273C0D85EB10-1.json.xz
Is64Bit True
IsPAE   False
layer_name      0 WindowsIntel32e
memory_layer    1 WindowsCrashDump64Layer
base_layer      2 FileLayer
KdVersionBlock  0xf80323a0f408
Major/Minor     15.19041
MachineType     34404
KeNumberProcessors      4
SystemTime      2026-03-25 02:06:06+00:00
NtSystemRoot    C:\Windows
NtProductType   NtProductWinNt
NtMajorVersion  10
NtMinorVersion  0
PE MajorOperatingSystemVersion  10
PE MinorOperatingSystemVersion  0
PE Machine      34404
PE TimeDateStamp        Thu Apr 17 15:14:37 2070
```

Fig 4.1 Memory Analysis

```
C:\Users\student\volatility3-stable> python vol.py -f MA202-55-20260325-020603.dmp windows.pslist
Volatility 3 Framework 2.27.0
Progress:  100.00               PDB scanning finished
PID     PPID    ImageFileName   Offset(V)       Threads Handles SessionId       Wow64   CreateTime      ExitTime        File output

4       0       System  0xb2815294080   197     -       N/A     False   2026-03-25 01:49:18.000000 UTC  N/A     Disabled
108     4       Registry        0xb28152430080  4       -       N/A     False   2026-03-25 01:49:26.000000 UTC  N/A     Disabled
436     4       smss.exe        0xb2815608f040  2       -       N/A     False   2026-03-25 01:49:28.000000 UTC  N/A     Disabled
640     552     csrss.exe       0xb28158772140  12      -       0       False   2026-03-25 01:49:30.000000 UTC  N/A     Disabled
728     552     wininit.exe     0xb28159e79080  1       -       0       False   2026-03-25 01:49:30.000000 UTC  N/A     Disabled
736     720     csrss.exe       0xb28159ec4140  13      -       1       False   2026-03-25 01:49:30.000000 UTC  N/A     Disabled
832     720     winlogon.exe    0xb28159f781c0  3       -       1       False   2026-03-25 01:49:31.000000 UTC  N/A     Disabled
840     728     services.exe    0xb28159f77080  7       -       0       False   2026-03-25 01:49:31.000000 UTC  N/A     Disabled
924     728     lsass.exe       0xb2815a6b6080  13      -       0       False   2026-03-25 01:49:34.000000 UTC  N/A     Disabled
388     840     svchost.exe     0xb2815a69c8c0  15      -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
488     728     fontdrvhost.ex  0xb2815a726080  5       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
500     832     fontdrvhost.ex  0xb2815a727080  5       -       1       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
600     840     svchost.exe     0xb2815a743240  13      -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
1068    840     svchost.exe     0xb2815a7e70c0  6       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
1136    840     svchost.exe     0xb2815a866080  4       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
1144    840     svchost.exe     0xb2815a869080  25      -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
1300    840     svchost.exe     0xb2815a8830c0  3       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
1308    840     svchost.exe     0xb2815a88b080  3       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
1316    840     svchost.exe     0xb2815a8e0080  4       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
1324    840     svchost.exe     0xb2815a8ab080  2       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
1332    840     svchost.exe     0xb2815a8b50c0  2       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
1340    840     svchost.exe     0xb2815a8b6080  2       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
1348    840     svchost.exe     0xb2815a8b9080  3       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
1356    840     svchost.exe     0xb2815a8bc080  7       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
1504    832     dwm.exe 0xb2815a947080  14      -       1       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
1548    840     svchost.exe     0xb2815a936080  5       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
1596    840     svchost.exe     0xb2815a937080  6       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
1632    840     svchost.exe     0xb2815a9ab080  7       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
1696    840     IntelCpHDCPSvc  0xb2815a9ae080  3       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
1712    840     svchost.exe     0xb2815a9af080  15      -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
1768    840     svchost.exe     0xb2815a9e5080  11      -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
1880    840     svchost.exe     0xb2815aa4c080  7       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
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2020    840     svchost.exe     0xb2815aaf1080  1       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
2064    840     svchost.exe     0xb2815ab27080  10      -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
2108    840     svchost.exe     0xb2815ab3c080  6       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
2184    840     IntelCpHeciSvc  0xb2815abad080  3       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
2240    840     svchost.exe     0xb2815abe8080  2       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
2256    840     svchost.exe     0xb2815abe9080  4       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
2264    840     svchost.exe     0xb2815abe7080  13      -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
2416    840     svchost.exe     0xb2815ac91080  5       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
2504    840     igfxCUIService  0xb2815acb2080  2       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
2568    840     svchost.exe     0xb2815ad130c0  8       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
2700    840     svchost.exe     0xb2815ae22080  2       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
2740    840     svchost.exe     0xb2815ae34080  5       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
2764    840     svchost.exe     0xb2815ae6c080  3       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
2772    840     svchost.exe     0xb2815ae880c0  3       -       0       False   2026-03-25 01:49:35.000000 UTC  N/A     Disabled
```

Fig 4.2 Process Analysis

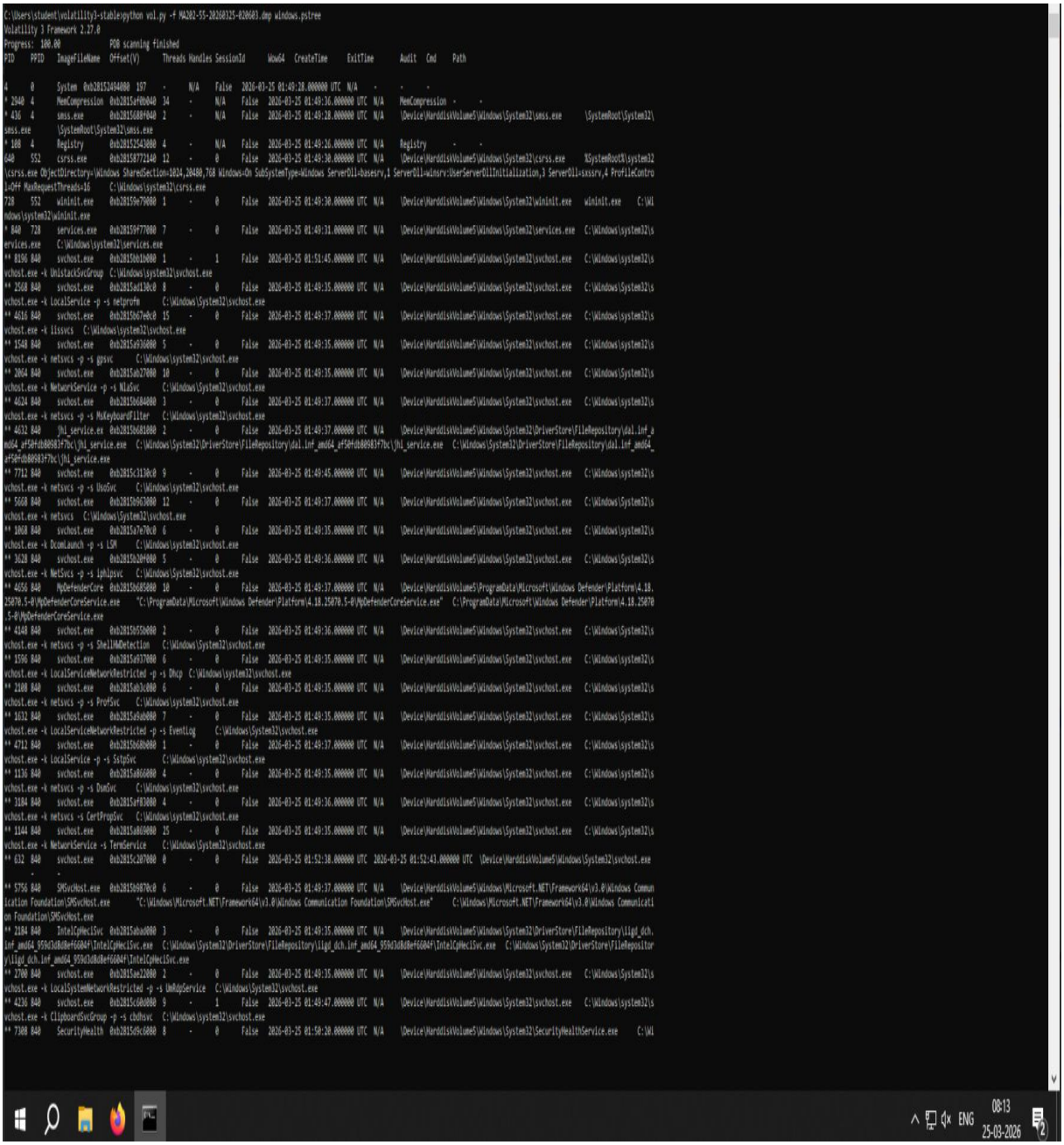

Fig 4.3 Parent child relationship of process

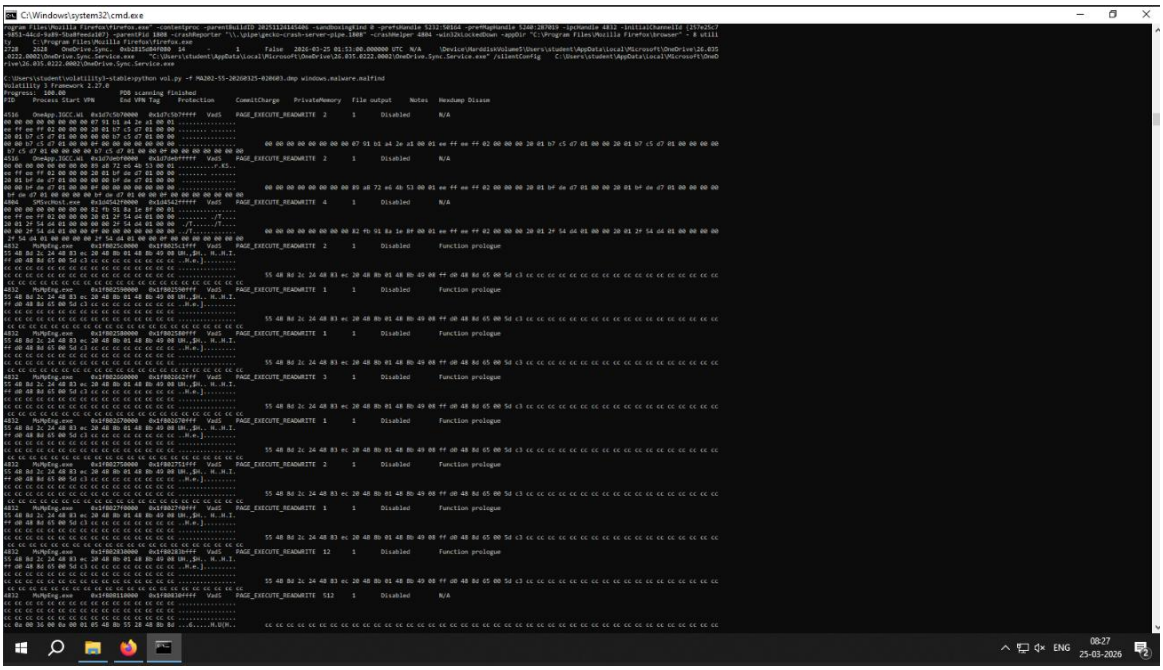

Fig 4.4 Malware Detection

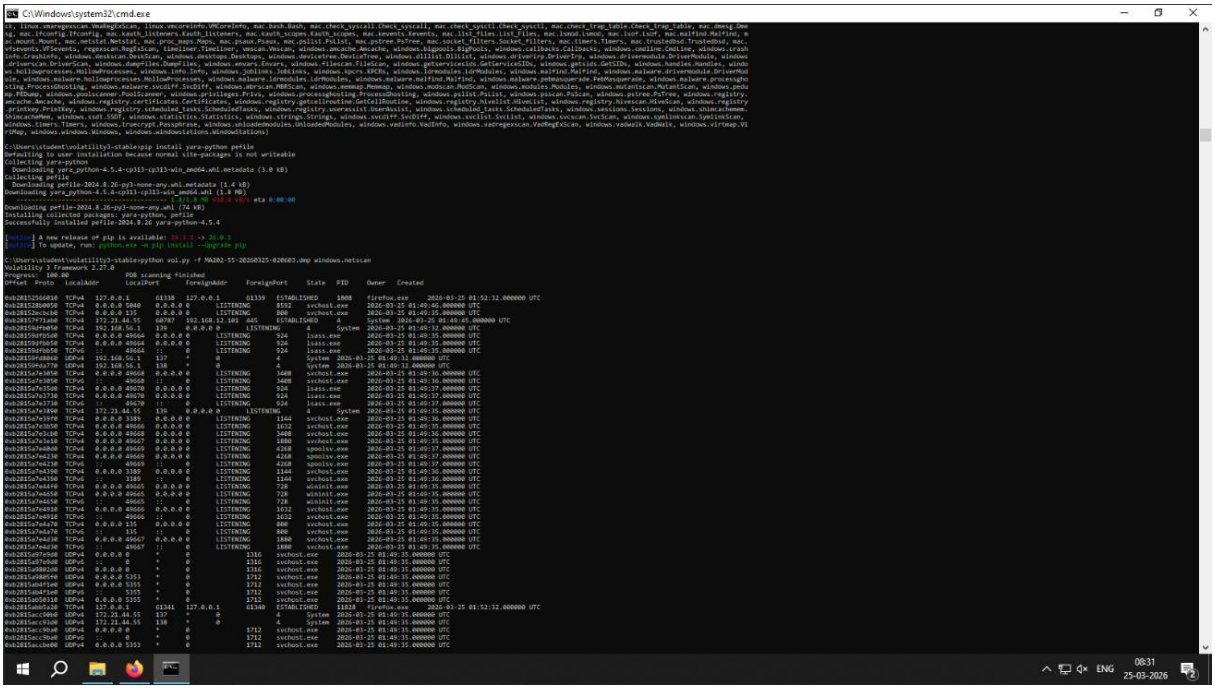

Fig 4.5 Network Analysis

CHAPTER 5

CONCLUSION AND FEATURE ENHACEMENT

5.1 OVERALL ANALYSIS OF PROJECT VIABLITIES

The Memory Acquisition and Analysis Toolkit is found to be highly viable based on technical, economic, and operational aspects. From a technical perspective, the system is feasible as it utilizes existing and well-established tools such as DumpIt, WinPmem, Volatility, and Rekall, which are compatible with modern operating systems. The implementation does not require complex algorithms or specialized hardware, making it easy to develop and deploy using current technology.

From an economic perspective, the project is cost-effective as it mainly relies on open-source and freely available tools. The hardware requirements are minimal and commonly available, reducing the overall cost of implementation. The estimated effort and cost remain within acceptable limits, making the project suitable for academic and practical applications without requiring significant financial investment.

From an operational perspective, the system is user-friendly and follows a structured workflow, making it easy to operate and manage. It supports effective memory acquisition and analysis, helping users identify security threats and perform forensic investigations efficiently. Although basic technical knowledge is required, the system simplifies complex processes through an organized approach.

5.2 PROBLEM ENCOUNTERED AND POSSIBLE SOLUTIONS

During the development of the Memory Acquisition and Analysis Toolkit, several challenges were encountered at different stages of the project. One of the major problems was related to memory acquisition, where improper tool execution or insufficient permissions sometimes resulted in incomplete or corrupted memory dump files. This issue was resolved by ensuring that the tools were executed with administrator privileges and by verifying the integrity of the

generated dump files.

Another challenge was data transfer between Windows and Kali Linux environments. Issues such as shared folder configuration errors, network connectivity problems, and file visibility difficulties made it hard to transfer large memory dump files. This problem was addressed by using alternative methods such as setting up a local HTTP server, configuring proper network settings, and verifying correct file paths to ensure smooth transfer of data.

The project also faced difficulties during the analysis phase, where incorrect commands or incompatible profiles in forensic tools like Volatility caused errors in extracting information. This was solved by carefully selecting the correct profiles, verifying tool compatibility, and referring to proper documentation. Additionally, handling large memory dump files required significant system resources, which sometimes affected performance.

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5.3 SUMMARY OF PROJECT WORK

The Memory Acquisition and Analysis Toolkit project focuses on capturing and analyzing live system memory (RAM) to support digital forensic investigations. The project was developed using a structured approach that includes memory acquisition from a Windows system, secure transfer of the memory dump, and detailed analysis in a Kali Linux environment using tools such as Volatility and Rekall. The system successfully demonstrates how volatile memory data can be preserved and analyzed to extract important information such as running processes, network connections, and potential security threats.

Throughout the project, various phases such as planning, design, implementation, testing, and documentation were completed systematically. Diagrams like ER, DFD, UML, and system architecture were used to design and understand the system workflow. The project also addressed challenges related to memory acquisition, data transfer, and analysis, and provided

effective solutions to overcome them.

Overall, the project achieves its objective of providing a reliable and efficient method for memory forensics. It enhances the ability to detect hidden or fileless malware and improves the accuracy of digital investigations. The toolkit provides a practical and cost-effective solution, making it suitable for both academic purposes and real-world cybersecurity applications.

5.4 LIMITATIONS

The Memory Acquisition and Analysis Toolkit has certain limitations despite its effectiveness in memory forensics. One of the primary limitations is that the system focuses only on volatile memory (RAM) analysis and does not include full disk forensics or recovery of deleted files from storage devices. As a result, it may not provide a complete investigation when persistent data analysis is required.

Another limitation is that the system relies on external tools such as DumpIt, WinPmem, Volatility, and Rekall for memory acquisition and analysis. Any limitations or compatibility issues in these tools can affect the overall performance of the system.

The system also requires technical knowledge to operate effectively. Users must be familiar with command-line tools and forensic concepts to correctly acquire memory and interpret analysis results. Handling large memory dump files can be resource-intensive, requiring sufficient RAM and storage, which may affect system performance.

Furthermore, the toolkit does not provide real-time monitoring or automated threat prevention, as it is designed for post-acquisition forensic analysis. The accuracy of results depends on the quality of the memory dump and proper usage of analysis tools. These limitations highlight areas for improvement in future enhancements of the system.

5.5 FUTURE ENHANCEMENT

The Memory Acquisition and Analysis Toolkit can be further improved by adding several advanced features to enhance its functionality and usability. One possible enhancement is the integration of real-time memory monitoring, which would allow the system to continuously observe system activities and detect threats as they occur, rather than only performing post-analysis. This would improve the system's capability in handling active cyber threats.

Another enhancement could be the development of a graphical user interface (GUI) to make the system more user-friendly. Currently, many operations are performed using command-line tools, which may be difficult for beginners. A GUI-based system would simplify the process of memory acquisition, analysis, and report generation, making it accessible to a wider range of users.

Another enhancement could be the development of a graphical user interface (GUI) to make the system more user-friendly. Currently, many operations are performed using command-line tools, which may be difficult for beginners. A GUI-based system would simplify the process of memory acquisition, analysis, and report generation, making it accessible to a wider range of users.

The system can also be extended to support multiple operating systems such as Linux and macOS for memory acquisition, increasing its applicability in diverse environments. Additionally, incorporating automation features can reduce manual effort by automatically executing analysis commands and generating reports.

5.6 PROJECT OUTCOMES AND DISCUSSION

The Memory Acquisition and Analysis Toolkit successfully achieves its objective of capturing and analyzing live system memory for digital forensic investigations. The system is able to acquire memory data from a running Windows system without shutting it down, ensuring that important volatile information is preserved. The acquired memory dump is effectively transferred to a Kali Linux environment, where it is analyzed using tools such as Volatility and Rekall. The results include detailed information about running processes, network connections, and system activities.

The outcomes of the project demonstrate that memory forensics plays a crucial role in identifying hidden threats, especially fileless malware that cannot be detected through traditional disk-based analysis. The system provides accurate and meaningful results, which can be used as evidence in forensic investigations. The structured workflow implemented in the project improves the efficiency and reliability of the entire process, from acquisition to report generation.

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Consent Letter

We,**PROF.PARTH PARMAR,VATSAL KAKKAD,HEMANSHI JALU**,hereby give our full consent and authorization for filling a patent publication application for for the project titled **"Live Memory Acquisition And Analysis Toolkit"**.

We hereby authorize Marwadi University and its legal representation to file the patent publication application and act on our behalf regarding any matters related to this filling.

Date:

Name:PROF. PARTH PARMAR

Signature:

Live Memory Acquisition And Analysis Toolkit

Abstract

The Memory Acquisition and Analysis Toolkit is a digital forensic system designed to capture, transfer, and analyze live system memory (RAM) to support cybersecurity investigations. RAM contains important volatile data such as running processes, active network connections, user sessions, and traces of malicious activities, which are lost once the system is shut down. Therefore, capturing this data in real time is essential for effective forensic analysis. The system focuses on acquiring memory from a Windows-based environment using tools like DumpIt or WinPmem, allowing investigators to collect data without interrupting system operations. The captured memory dump is then transferred to a Kali Linux environment for analysis using forensic frameworks such as Volatility and Rekall. The main objective of this system is to provide a structured and reliable approach for extracting useful information from memory, including system activities and potential malicious artifacts. The toolkit helps identify suspicious behavior, detect hidden or fileless malware, and analyze evidence not found on the hard disk. It also ensures an organized workflow for memory acquisition, analysis, and report generation, improving the efficiency and accuracy of digital forensic investigations.

Student 1 Signature:

Student 2 Signature:

Internal Guide Name and Signature:

INVENTION DISCLOSURE FORM FOR PATENTS

1. Particulars of Inventors

Mr./Ms/ Dr.	Name (Full)	Department	Designation	Mobile No.	Email	Postal Address
Mr	Vatsal Kakkad	Computer Engineering	Student	+91989 8709609	Vatsal.kakkad125750@marwadiuniversity.ac.in	Marwadi University
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2. Provide title of the Invention:

Live Memory Acquisition and Analysis Toolkit

3.In 100 words or less, please provide an abstract or summary of the invention:

The invention relates to a Memory Acquisition and Analysis Toolkit designed for digital forensic investigations. The system enables live capture of volatile memory (RAM) from a running system without shutdown and performs structured analysis using forensic frameworks. It extracts critical information such as running processes, network connections, and hidden or fileless malware. The toolkit provides a systematic workflow for memory acquisition, secure transfer, analysis, and report generation. This invention improves the accuracy and efficiency of cybercrime investigations by preserving volatile evidence and detecting advanced threats that are not identifiable through traditional disk-based forensic methods.

4. Detail description of the invention:(Answer to all below are required in detail)

a.Problem the invention is solving

The invention solves the problem of missing important evidence during digital investigations by focusing on live memory (RAM), which is usually lost when a system is turned off. Traditional methods mainly analyze stored data and often fail to detect hidden activities like fileless malware, running processes, or active network connections. It also addresses the difficulty of using multiple separate tools, which can be complex and time-consuming. This toolkit provides a simple and organized way to capture and analyze memory data without

shutting down the system, helping investigators easily find critical information and improving the accuracy and efficiency of cybercrime analysis.

b.General Utility/application of the invention

The invention can be widely used in the field of digital forensics and cybersecurity to support the investigation of cybercrimes and security incidents. It helps forensic analysts, security professionals, and law enforcement agencies to capture and analyze live system memory in order to identify suspicious activities, detect hidden or fileless malware, and recover critical information such as running processes and network connections. The toolkit is also useful in incident response, system monitoring, and security auditing, where quick and accurate analysis of volatile data is required. Overall, it improves the ability to investigate, detect, and respond to modern cyber threats effectively.

c.Advantages of the invention disclosing about the increased efficiency/efficacy

The invention offers significant advantages by improving the efficiency and effectiveness of digital forensic investigations. It enables live memory acquisition without shutting down the system, ensuring that critical volatile data is preserved and analyzed in real time. The structured workflow reduces manual effort and simplifies the process, allowing faster and more accurate analysis. It enhances the ability to detect hidden and fileless malware that traditional methods often miss. By integrating acquisition, transfer, and analysis steps, the system minimizes errors and improves reliability. Overall, the invention increases investigation speed, accuracy, and the quality of forensic evidence, making cyber threat detection more efficient.

d.Best way of using the invention as well as possible variants

The invention is best used by first performing live memory acquisition on a running system using a suitable tool, ensuring that the system is not shut down so that volatile data is preserved. The captured memory dump is then securely transferred to an analysis environment, where forensic tools are used to extract and examine information such as processes, network activity, and potential threats. The results are interpreted and compiled into a report for investigation purposes. Possible variants of the invention include integrating automated analysis features, developing a graphical user interface for ease of use, supporting multiple operating systems for memory acquisition, and combining memory analysis with disk and network forensics to provide a more comprehensive investigation approach.

e.Working of invention along with Drawing, schematics and flow diagrams if required with complete explanations

The working of the invention follows a structured sequence of steps for capturing and analyzing live system memory. Initially, the system performs **live memory acquisition** on a running machine using a memory acquisition tool, which creates a complete memory dump file without shutting down the system. This ensures that all volatile data such as running processes, network connections, and temporary system information is preserved. The acquired memory dump is then securely transferred to a separate analysis environment to maintain integrity and avoid contamination of evidence.

In the next stage, the memory dump file is processed using forensic analysis tools that extract useful information from the raw data. These tools identify system activities, active processes, loaded modules, and potential malicious artifacts. The extracted data is then organized and interpreted to detect suspicious behavior or hidden threats such as fileless malware. Finally, the analyzed results are compiled into a structured report for further investigation and documentation.

5. Have you conducted Primary Patent Search? Yes / No (if yes, attach the patent search results)

No

6. Existing state-of-the-art and prior arts: (Brief background of the existing knowledge/product/process in the market)

The current state-of-the-art in digital forensics primarily includes memory analysis tools such as Volatility and Rekall, along with commercial solutions like Redline and Magnet AXIOM. These tools are capable of extracting useful information such as processes, network connections, and system artifacts from memory dump files. Volatility, in particular, is widely used due to its modular and cross-platform capabilities for analyzing RAM data.However, existing solutions often require high technical expertise, involve multiple manual steps, and lack an integrated workflow for acquisition, transfer, and analysis. Additionally, some tools like Rekail are no longer actively maintained, limiting their future usability.Therefore, there is a need for a more structured, user-friendly, and efficient system that simplifies the overall memory forensic process.

7. List out the known ways about how others have tried to solve the same or similar problems? Indicate the disadvantages of these approaches. In addition, please identify any prior art documentation or other material that explains or provides examples of such prior art efforts.

S. No.	Existing state of art	Existing state of art	Overcome (How Your Invention Solves It)
1	Overcome (How Your Invention Solves It)	Requires strong technical knowledge and manual command usage	Provides a structured workflow, making analysis easier and more organized
2	Rekall Framework	Limited updates and reduced support	Uses a flexible approach that can integrate with updated tools
3	Redline (Memory Analysis Tool)	Limited customization and mostly GUI-based with restricted control	Combines flexibility of tools with a systematic process for better control

4	Magnet AXIOM	Commercial tool with high cost	Uses open-source tools, making it cost-effective
5	Traditional Disk Forensics	Cannot capture volatile memory data	Focuses on live memory acquisition to preserve volatile data
6	Manual Memory Analysis Approach	Time-consuming and error-prone	Automates and organizes the process to improve efficiency and accuracy
7	Individual Tools Used Separately	Lack of integration and workflow management	Integrates acquisition, transfer, and analysis into a single structured system

8. List the Technical features and Elements of the invention along with the Description of your invention from start to end.

A. Live Memory Acquisition Module:
The system captures volatile memory (RAM) from a running system without shutting it down, ensuring that critical real-time data such as processes and network activity is preserved.

B. Cross-Platform Environment Setup:
The invention uses a Windows environment for memory acquisition and a Kali Linux environment for analysis, providing flexibility and efficient processing.

C.Secure Data Transfer Mechanism:
structured method is used to transfer large memory dump files securely between systems, ensuring data integrity and preventing data loss.

D. Memory Analysis Framework Integration:
The system integrates forensic tools to analyze memory dumps and extract useful information such as running processes, network connections, and system activities.

E. Process and Network Analysis:
The system identifies active processes and network connections, helping in detecting suspicious or unauthorized activities.

F. Malware Detection Capability:
The invention supports detection of hidden and fileless malware that resides only in memory

and cannot be detected using traditional disk-based methods.

G. Structured Workflow Management:

The system follows a step-by-step process including acquisition, transfer, analysis, and reporting, making the investigation process organized and efficient.

H. Report Generation Module:
The system generates structured reports containing analysis results, making it easier for investigators to understand and use the findings.

9.List out the features of your invention which are believed to be new and distinguish them over the closest technology.

The invention introduces several novel features that distinguish it from existing memory forensic solutions. One of the key innovations is the integration of live memory acquisition, secure transfer, analysis, and reporting into a single structured workflow, whereas existing tools typically perform these functions separately and require manual coordination. This unified approach improves efficiency, reduces complexity, and minimizes the chances of error.

Another distinguishing feature is the system's focus on live memory acquisition without shutting down the system, ensuring that critical volatile data is preserved. While existing tools support memory analysis, they often lack an organized end-to-end process that ensures proper handling of data from acquisition to final reporting. The invention also enhances usability by providing a simplified and systematic process, making it more accessible compared to traditional command-line-based approaches that require high technical expertise.

10.Has the invention been built or tested or implemented? If yes please provide the Efficiency/Efficacy details of the invention

No

11. Briefly state when and how you first conceived this idea?

The idea for this invention was first conceived during academic study and practical exposure to digital forensics, when it was observed that traditional forensic methods were unable to capture important volatile data stored in system memory. While working on cybersecurity-related tasks and tools, it became evident that there was a gap in having a simple and structured approach for live memory acquisition and analysis. This led to the concept of developing a toolkit that could integrate the entire process—from capturing memory to analyzing and generating reports—into a single, organized workflow, making forensic investigations more efficient and reliable.

12. Have you sold, offered for sale, publicly used or published anything related to this invention? If yes, please briefly explain the dates and circumstances. List those individuals to whom you have revealed your invention. Were non-discloser documents signed prior to discloser in each case? Please state any deadlines of which you may be aware for filing an application on this

invention.

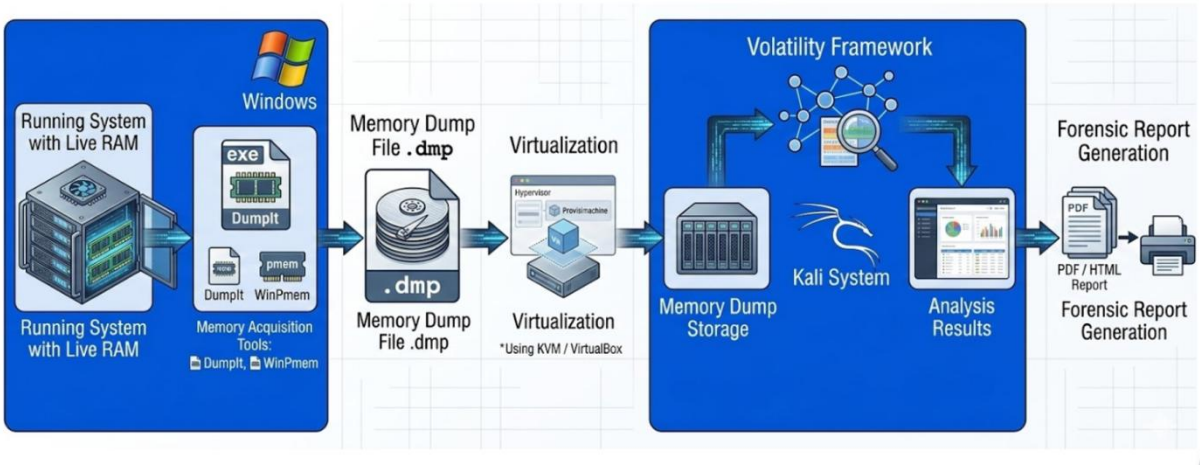
No

13.Include any reasons that your invention would not have been obvious to someone of average skill in the art.

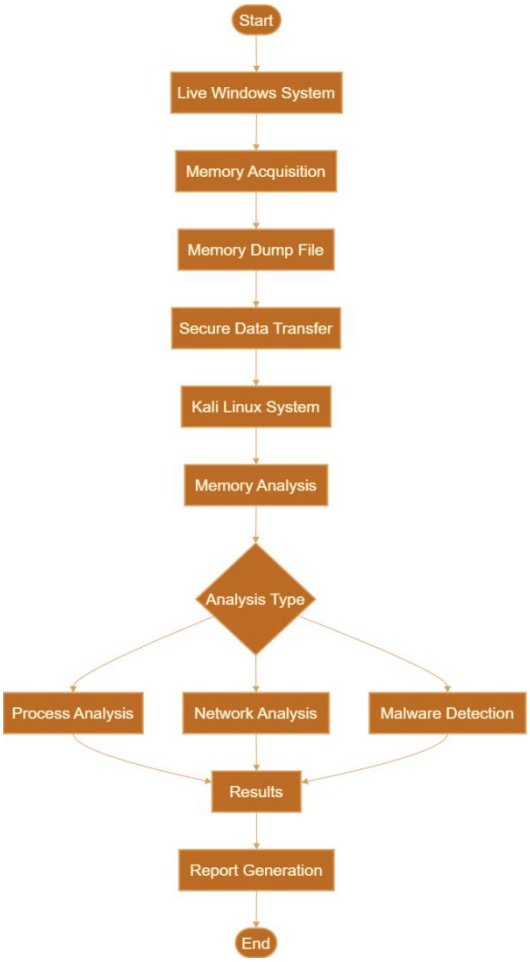
The invention would not have been obvious to a person of average skill in the field because, although individual tools for memory acquisition and analysis already exist, they are typically used separately and require significant manual effort and expertise to integrate. The novelty lies in combining these independent processes into a single, structured, and efficient workflow that simplifies live memory forensics from acquisition to report generation. Additionally, the focus on preserving volatile data through live acquisition and systematically detecting advanced threats such as fileless malware enhances its practical value. This integrated, user-oriented approach goes beyond routine use of existing tools and provides a more effective and organized solution that is not directly suggested by current technologies.

14.Drawings/Flowchart/Table

Architecture



Flowchart


Start
Live Windows System
Memory Acquisition
Memory Dump File
Secure Data Transfer
Kali Linux System
Memory Analysis
Analysis Type
Process Analysis
Network Analysis
Malware Detection
Results
Report Generation
End

Project Outcome Documents

The Memory analysis has important applications in the field of digital forensics and cybersecurity. It helps investigators examine the live memory (RAM) of a system to identify ongoing activities such as running processes, active network connections, and system behavior. This is especially useful in cybercrime investigations, where attackers may use techniques that do not leave traces on the hard disk. By analyzing memory, investigators can uncover hidden or suspicious activities and gather valuable evidence that would otherwise be lost when the system is turned off.

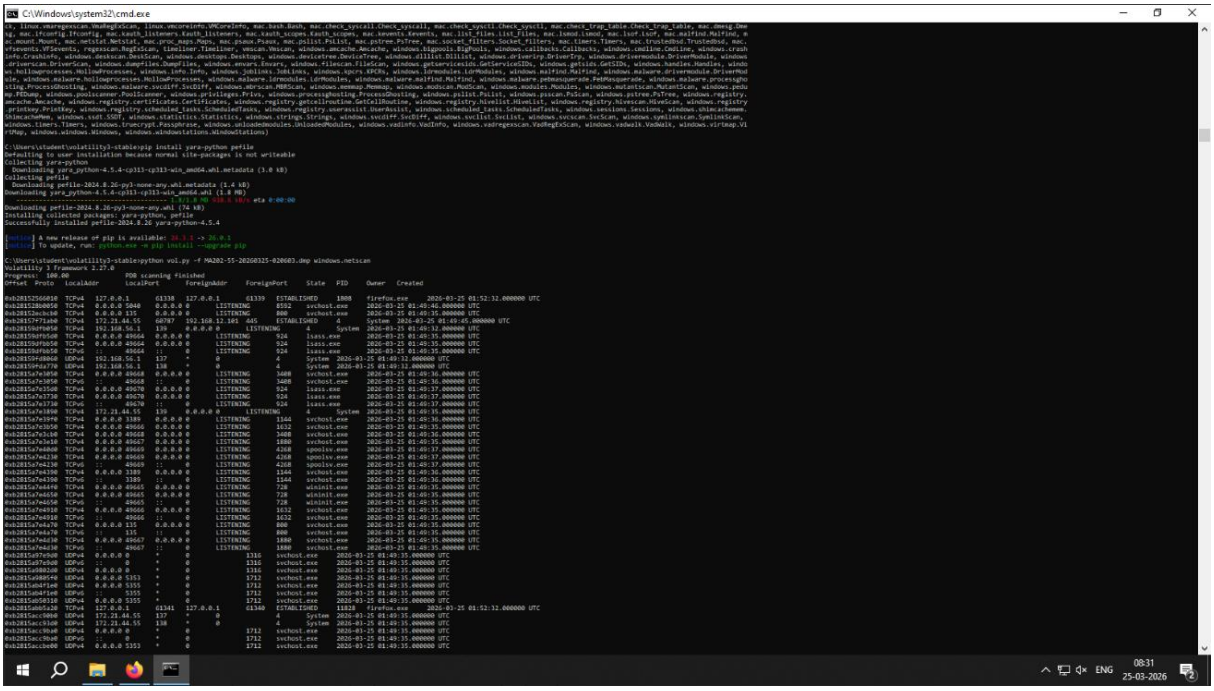
In addition, memory analysis is widely used in incident response and system security monitoring. Security professionals use it to quickly detect threats like fileless malware, unauthorized access, and in-memory attacks. It also helps organizations understand how a system was compromised and take necessary actions to prevent future attacks. Overall, memory analysis improves the accuracy and speed of threat detection, making it a powerful tool in modern cybersecurity practices.

```
C:\Users\student\volatility3-stable>python vol.py -f MA202-S5-20260325-020603.dmp windows.info
Volatility 3 Framework 2.27.0
Progress:  100.00               PDB scanning finished
Variable        Value

Kernel Base     0xf80322e00000
DTB     0x1ad000
Symbols file:///C:/Users/student/volatility3-stable/volatility3/symbols/windows/ntkrnlmp.pdb/B6121DA15DDCF625C8C7273C0085EB10-1.json.xz
Is64Bit True
IsPAE   False
layer_name      0 WindowsIntel32e
memory_layer    1 WindowsCrashDump64Layer
base_layer      2 FileLayer
KdVersionBlock  0xf80323a0f408
Major/Minor     15.19041
MachineType     34404
KeNumberProcessors      4
SystemTime      2026-03-25 02:06:06+00:00
NtSystemRoot    C:\Windows
NtProductType   NtProductWinNt
NtMajorVersion  10
NtMinorVersion  0
PE MajorOperatingSystemVersion  10
PE MinorOperatingSystemVersion  0
PE Machine      34404
PE TimeDateStamp        Thu Apr 17 15:14:37 2070
```

Network analysis plays a crucial role in cybersecurity and digital forensics by helping investigators monitor and examine communication between systems. It allows the identification of active network connections, open ports, and data transfers occurring within a system. This is useful for detecting suspicious activities such as unauthorized access, data exfiltration, or communication with malicious servers. By analyzing network-related data from memory, investigators can trace attacker behavior and understand how a system is interacting with external entities.

In addition, network analysis is widely used in incident response and security monitoring to detect and prevent cyber threats. It helps security professionals identify abnormal traffic patterns, potential intrusions, and compromised systems within a network. This information is essential for taking quick action to block threats and secure the system. Overall, network analysis enhances the ability to detect, investigate, and respond to cyber incidents effectively, making it an important component of modern cybersecurity.


C:\Windows\system32\cmd.exe
C:\Users\student\volatility3-stable>pip install yara-python pefile
Defaulting to user installation because normal site-packages is not writeable
Collecting yara-python
Downloading yara_python-4.5.4-cp313-cp313-win_amd64.whl.metadata (3.0 kB)
Collecting pefile
Downloading pefile-2024.8.26-py3-none-any.whl.metadata (1.4 kB)
Downloading yara_python-4.5.4-cp313-cp313-win_amd64.whl (1.8 MB)
Downloading pefile-2024.8.26-py3-none-any.whl (74 kB)
Installing collected packages: yara-python, pefile
Successfully installed pefile-2024.8.26 yara-python-4.5.4
[notice] A new release of pip is available: 24.3.1 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
C:\Users\student\volatility3-stable>python vol.py -f MA202-55-20260325-020603.dmp windows.netscan
Volatility 3 Framework 2.27.0
Progress: 100.00 PDB scanning finished
Offset Proto LocalAddr LocalPort ForeignAddr ForeignPort State PID Owner Created
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0xb2815280b050 TCPv4 0.0.0.0 5040 0.0.0.0 0 LISTENING 8592 svchost.exe 2026-03-25 01:49:46.000000 UTC
0xb2815ac3c040 TCPv4 0.0.0.0 135 0.0.0.0 0 LISTENING 800 svchost.exe 2026-03-25 01:49:35.000000 UTC
0xb2815f71a040 TCPv4 172.21.44.55 60797 192.168.12.101 445 ESTABLISHED 4 System 2026-03-25 01:49:45.000000 UTC
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0xb2815f9f05d0 TCPv4 0.0.0.0 49664 0.0.0.0 0 LISTENING 924 lsass.exe 2026-03-25 01:49:35.000000 UTC
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0xb2815f4f8b60 UDPv4 192.168.56.1 137 * 0 4 System 2026-03-25 01:49:32.000000 UTC
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0xb2815e7a1050 TCPv4 0.0.0.0 49668 0.0.0.0 0 LISTENING 3608 svchost.exe 2026-03-25 01:49:36.000000 UTC
0xb2815e7a1050 TCPv6 :: 49668 :: 0 LISTENING 3608 svchost.exe 2026-03-25 01:49:36.000000 UTC
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0xb2815e7e4860 TCPv4 0.0.0.0 49669 0.0.0.0 0 LISTENING 4268 spoolsv.exe 2026-03-25 01:49:37.000000 UTC
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0xb2815e7e4910 TCPv4 0.0.0.0 49666 0.0.0.0 0 LISTENING 1632 svchost.exe 2026-03-25 01:49:35.000000 UTC
0xb2815e7e4910 TCPv6 :: 49666 :: 0 LISTENING 1632 svchost.exe 2026-03-25 01:49:35.000000 UTC
0xb2815e7e4a70 TCPv4 0.0.0.0 135 0.0.0.0 0 LISTENING 800 svchost.exe 2026-03-25 01:49:35.000000 UTC
0xb2815e7e4a70 TCPv6 :: 135 :: 0 LISTENING 800 svchost.exe 2026-03-25 01:49:35.000000 UTC
0xb2815e7e4d40 TCPv4 0.0.0.0 49667 0.0.0.0 0 LISTENING 1880 svchost.exe 2026-03-25 01:49:35.000000 UTC
0xb2815e7e4d40 TCPv6 :: 49667 :: 0 LISTENING 1880 svchost.exe 2026-03-25 01:49:35.000000 UTC
0xb2815e97e9d0 UDPv4 0.0.0.0 0 * 0 1316 svchost.exe 2026-03-25 01:49:35.000000 UTC
0xb2815e97e9d0 UDPv6 :: 0 * 0 1316 svchost.exe 2026-03-25 01:49:35.000000 UTC
0xb2815e980240 UDPv4 0.0.0.0 0 * 0 1316 svchost.exe 2026-03-25 01:49:35.000000 UTC
0xb2815e980f40 UDPv4 0.0.0.0 5353 * 0 1712 svchost.exe 2026-03-25 01:49:35.000000 UTC
0xb2815e9d1e40 UDPv4 0.0.0.0 5355 * 0 1712 svchost.exe 2026-03-25 01:49:35.000000 UTC
0xb2815e9d1e40 UDPv6 :: 5355 * 0 1712 svchost.exe 2026-03-25 01:49:35.000000 UTC
0xb2815e9d810d UDPv4 0.0.0.0 5355 * 0 1712 svchost.exe 2026-03-25 01:49:35.000000 UTC
0xb2815a065a20 TCPv4 127.0.0.1 61341 127.0.0.1 61340 ESTABLISHED 11828 firefox.exe 2026-03-25 01:52:32.000000 UTC
0xb2815acc90b0 UDPv4 172.21.44.55 137 * 0 4 System 2026-03-25 01:49:35.000000 UTC
0xb2815acc91d0 UDPv4 172.21.44.55 138 * 0 4 System 2026-03-25 01:49:35.000000 UTC
0xb2815acc9ba0 UDPv4 0.0.0.0 0 * 0 1712 svchost.exe 2026-03-25 01:49:35.000000 UTC
0xb2815acc9ba0 UDPv6 :: 0 * 0 1712 svchost.exe 2026-03-25 01:49:35.000000 UTC
0xb2815acc9dc0 UDPv4 0.0.0.0 5353 * 0 1712 svchost.exe 2026-03-25 01:49:35.000000 UTC
ENG
08:31
25-03-2026